Predicting a Decline in Patient-Reported Outcomes for Cancer Patients on Chemotherapy

UE24CS352A Machine Learning Mini-Project, Problem 8 | Team E-17: Sampreeth Jaladi (PES1UG24AM246), Sharvesh K (PES1UG24AM260) | Replication of Ostberg & Peterson, CS229 Spring 2020

1. Problem statement

Chemotherapy often worsens patients' physical well-being, captured by the PROMIS Global Physical Health (GPH) T-score (population mean 50, SD 10). The goal is to predict, **at the first chemotherapy dose and using only data available then**, which patients will report poor physical health during treatment, so clinicians can target supportive care. Two binary outcomes from the paper are modelled: **Drop**, on-treatment GPH falls by at least 10 points (1 SD) from baseline; and **Threshold**, on-treatment GPH is 40 or lower (1 SD below the mean). Models are compared by AUROC.

2. Dataset

The paper uses de-identified Stanford EHR data (STARR), which is not publicly available. We therefore wrote a **synthetic EHR generator** that produces *raw event-level* tables (patients, encounters, vitals, labs, diagnoses, procedures, medication classes, PROMIS surveys) with the same structure, so the paper's entire pipeline runs unchanged. A latent frailty variable drives both the observable EHR signals and the change in GPH, together with cancer type, regression to the mean of the baseline score, a non-linear age by albumin interaction and substantial unexplained noise. The generator also deliberately plants *post-chemotherapy* events that correlate with the outcome, to test for leakage, and charting errors in the vitals. Applying the paper's inclusion criteria to 1,715 simulated patients (cancer diagnosis from 2013, received chemotherapy, a baseline survey within 90 days before and an on-treatment survey 30-180 days after the first dose) gives **1,254 patients** (paper: 1,252). Outcome prevalence is Drop 18.5% and Threshold 22.6%. Absolute AUROCs therefore describe the simulator; the contribution is a faithful, tested re-implementation of the method.

3. Approach

Features (only events on or before the index date): vitals with physiologically impossible values removed; labs; procedures; diagnoses (full history, binary); medication classes; healthcare utilisation; demographics; and 6 baseline PROMIS features. Time-varying data from the last 180 days is summarised with the paper's exponential weight exp(-t/365). This gives 852 raw features (320 diagnosis, 336 procedure, 121 medication, 49 lab, 7 vital, 5 utilisation, 8 demographic, 6 PROMIS). **Preprocessing**: a 2.5% prevalence filter on labs, diagnoses and procedures, then labs handled by one of two strategies: (a) KNN imputation, or (b) tertile bins plus a missing indicator. **Models**: logistic regression (LASSO, Ridge, Elastic Net), random forest, LightGBM, a 3-layer MLP, an SVM (4 kernels) and KNN, plus a **soft-voting ensemble** that averages the probabilities of the tuned LASSO, Ridge, Elastic Net, random forest, LightGBM and Platt-scaled SVM. **Protocol**: a stratified 80/20 patient split; grid search with stratified 10-fold CV on the training set, scored by AUROC; each best model refitted on the training set and evaluated *once* on the test set, with 1,000-sample bootstrap 95% CIs.

4. Implementation

We wrote a Python package (`prodecline`) using scikit-learn and LightGBM with a staged CLI (data, train, ablate, figures). Training is checkpointed so an interrupted run resumes. **Every fitted preprocessing step** (prevalence filter, imputation, binning cut-points, one-hot categories, scaling) **sits inside the model pipeline**, so it is re-fitted on each training fold and never sees validation or test data. We also corrected two issues in the original study: it early-stopped LightGBM and the MLP on the *test* set, and it applied the prevalence filter to the whole cohort. Nineteen unit tests cover the pipeline; the key one deletes every post-index event and checks that the feature matrix is bit-identical, which proves that no future information is used.

5. Results

Model	Drop, KNN labs	Drop, binned labs	Threshold, KNN labs	Threshold, binned labs
LASSO	0.699	**0.730**	0.851	0.840
Ridge	0.638	0.676	0.841	0.843
Elastic Net	**0.717**	0.714	0.866	**0.868**
Random Forest	0.634	0.653	0.851	0.849
LightGBM	0.688	0.709	0.854	0.840
MLP	0.532	0.512	0.810	0.801
SVM	0.619	0.643	0.833	0.806
KNN	0.560	0.553	0.809	0.772
Ensemble (soft vote)	0.692	0.716	**0.868**	0.863

Table 1. Test-set AUROC (n = 251 held-out patients); best per column in bold. Mean CV AUROCs and bootstrap CIs are in `results/metrics.csv`.

Model selection. Ranked by cross-validated AUROC (the selection criterion), the ensemble was the top model in 4 of 4 settings (drop / knn labs 0.753, drop / binned labs 0.749, threshold / knn labs 0.821, threshold / binned labs 0.828). On the held-out test set it reached 0.692 (95% CI 0.602-0.776) for Drop and 0.868 (95% CI 0.816-0.915) for Threshold with KNN labs (paper: 0.771 and 0.752). The best single test scores were LASSO for Drop (0.730, binned labs) and Elastic Net for Threshold (0.868). Every gap between the ensemble and the best single model lies well inside the 95% CIs (about +/-0.09 for Drop with only 47 positive test patients), so the honest reading is that the ensemble and the penalised linear models perform equivalently, and the ensemble is the more robust choice by CV. As in the paper, MLP and KNN performed worst; both

struggle with several hundred mostly sparse features and ~1,000 training patients. KNN-imputed labs beat binned labs in 10 of 18 model and outcome pairs with overlapping CIs, so unlike the paper we find no clear winner between the two lab strategies.

Ablation (ensemble, KNN labs): with all features, Drop AUROC is 0.692. Removing the baseline PROMIS features lowers it to 0.559, and removing demographics and PROMIS together lowers it to 0.551 (paper: 0.770 to 0.599). Removing any other single group changes AUROC by less than the width of the CI. For Threshold the equivalent figures are 0.868 and 0.749. Baseline patient-reported status is the dominant predictor, which is clinically legitimate because it is routinely collected before treatment.

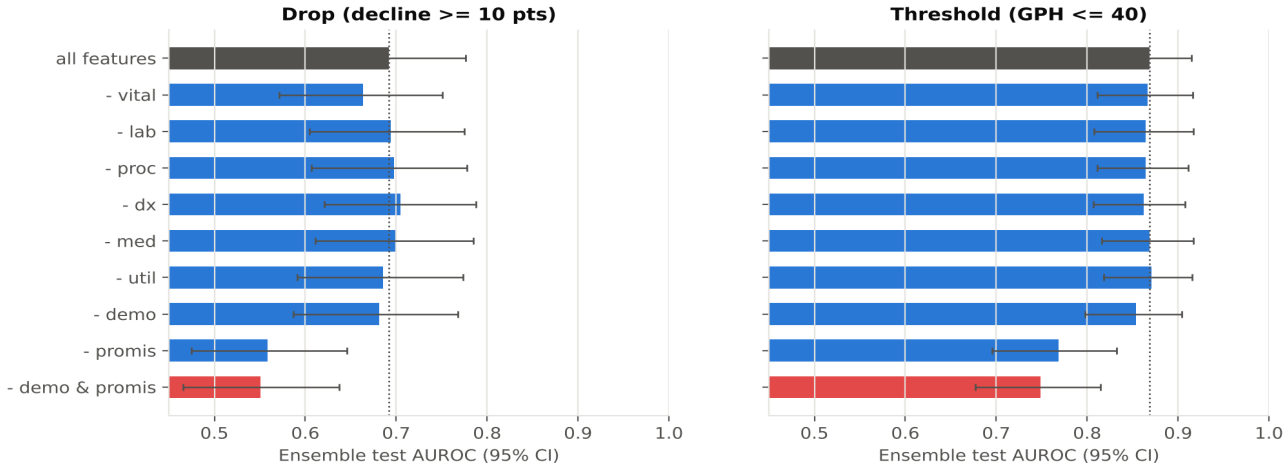


Figure 1. Ensemble test AUROC with each feature group removed (error bars: 95% bootstrap CI).

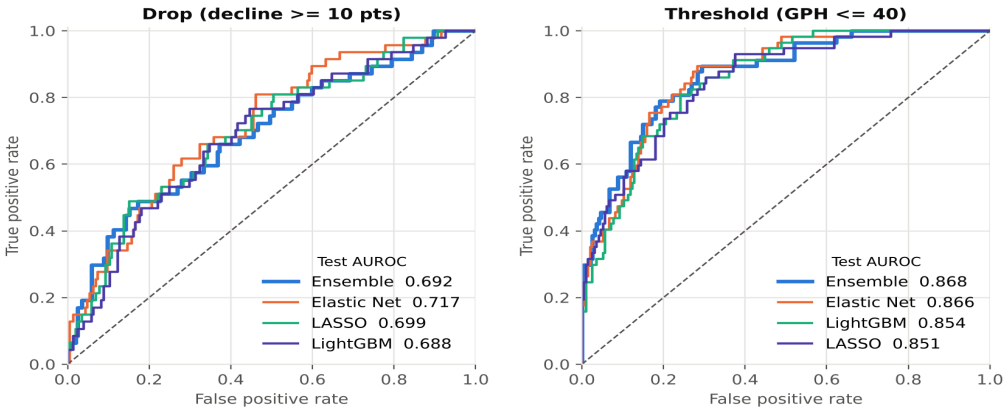


Figure 2. Test ROC curves: the ensemble and the three best single models (KNN-imputed labs).

6. Conclusions

We re-implemented the full Ostberg & Peterson pipeline, from cohort selection through time-weighted EHR featurisation, prevalence filtering, two lab strategies, eight tuned model families, a soft-voting ensemble and ablation, on a synthetic EHR built to mirror STARR. The paper's qualitative findings were reproduced: the ensemble is the most robust choice by cross-validation, with penalised linear models a close equal; KNN and the MLP underperform; and baseline PROMIS scores carry most of the signal, especially for Drop. Our protocol is stricter than the original (no test-set early stopping, preprocessing fitted inside CV, a regression test for leakage). **Limitations**: the data is simulated, so the AUROCs do not indicate clinical performance; there is a single train/test split; and the ensemble's CV score is mildly optimistic because its members were tuned on the same folds. **Future work**: validate on real EHR data (for example MIMIC-IV with a PRO proxy), use nested CV, check calibration, and weight the ensemble members by their CV performance.